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Frozen-Procedure Transfer in Battery-Free Event Service A Paired Digital Study with Censored Outcomes

Development manuscript

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Abstract

Battery-free event service requires powered observation and accurate encoded event times as well as energy-feasible delivery. We report a closed digital study with 73 logical cells and 77 actual attempts. On the original O0 development task, the candidate and a fixed rich-state control choose the same one-second program, share one response and pass eight events; a 60-second periodic control matches none. Three paired descendant lineages then compare fixed H0, descendants of fixed improver G0, and descendants of one model-proposed G1 procedure. All 30 effective selection cells pass before six methods are frozen. Across 27 final cells, H0 has 7 PASS, 1 submitted FAIL and 1 INCOMPLETE; G0 has 6/1/2; G1 has 8/0/1. Both reserved digital tasks pass in all 18 cells. The new O waveform separates event matching from service: both one-second submissions match eight events and pass energy checks but fail encoded-time tolerances, while three 0.75-second submissions pass. Four token-censored episodes retain their costs and denominators. The 8/9 G1 result is descriptive evidence conditional on one shared formation event, not stable recursive improvement or a unique mechanism; H0 also succeeds at 0.75 s. Physical qualification remains UNKNOWN and prior A/R product failures remain unaccepted.

1 Question and evidence scope

A battery-free event application must associate a change in external activity with sufficient powered observation, retained state and timely reception. Increasing the message count alone need not satisfy the required event semantics. Likewise, a successful event predicate cannot establish continuous sensing or discharge other reporting obligations. A transfer study therefore needs an unchanged task contract, matched hardware, a strong fixed event-state control, and separate records of the participant's judgment and the host's evaluation.

O0 first tests the extension of the existing A/R digital energy and receiver apparatus to vibration-based start, active keepalive and stop service. The participant receives the additional RMS source and declared sampling/wake costs rather than supplying an expected-event table. The comparison comprises a free legal candidate, an A-style periodic control and a fixed rich-state event control. It asks whether the candidate meets the original O service and whether an observed result exceeds the strong control. The closed development receipts support event-service feasibility within the nominal model, with no advantage over rich-state [1, 2].

The subsequent comparison asks whether a procedure that generates design methods transfers beyond its selection inputs. H denotes the procedure used by a fresh design episode; G denotes the procedure that proposes H descendants. We compare fixed H0, the descendants of fixed G0, and the descendants of one actually proposed G1. Six selected methods are frozen before any final input is delivered. Each consuming final episode sees the complete frozen current task facts, including its waveform. "Unexposed" therefore describes G/H formation and selection, not concealment of the task from its final designer.

The evidence comprises the complete actual-cell ledger, final per-event details, frozen selection and source-preservation records [8, 9, 7, 11]. The three descendant-lineage pairs are conditional on a single shared G1 formation event. They are not three independent G1 training runs. Similarly, eight transitions on one O waveform are not eight tasks, and cached responses are not repeated physical experiments. We retain submitted failures and unsubmitted censored episodes as different outcomes and include both in all reported denominators. H0 remains unchanged.

2 Original O service and matched digital apparatus

2.1 Unchanged external task and service predicate

All three O0 episodes use the original 21,600 s A0 supply schedule and zero initial capacitor voltage. The photovoltaic area is 15 cm^2 and capacitance is fixed at 15 mF. The six light intervals are 200, 0, 50, 200, 0 and 50 lux, at boundaries 0, 120, 150, 180, 300, 330 and 360 minutes. Their assumed electrical port powers are respectively 90, 0, 20, 90, 0 and 20 μW. The 3500 K light specification is metadata; lux is never converted to watts. Calibrated spectral irradiance and photovoltaic IV data are unavailable.

External RMS vibration is 0.08 g during 70–75, 130–135, 190–200 and 280–282 minutes and 0.005 g otherwise. Start requires continuously greater than 0.05 g for 3 s; stop requires continuously below 0.01 g for 60 s. The corresponding threshold equality resets confirmation. The event time is the confirmation instant. Every start and stop must reach the receiver within 10 s and have a timestamp error at most 1 s. Received active-state messages must cover each confirmed active interval with gaps at most 120 s. Darkness is included; idle false messages fail. A correct final state cannot cancel an earlier missing event or keepalive [4].

The four confirmed active intervals end at 4560, 8160, 12060 and 16980 s. They are service intervals, not a daily occupancy summary. In particular, the 130–135 minute activity window occurs within the first dark supply interval. The event predicate neither replaces event history with an hourly state nor accepts a retained NVM value as a newly observed event.

2.2 Inherited energy law and proposed RMS costs

The apparatus retains the A/R assumed source/cold-start/PMIC/capacitor and digital receiver parameters. Harvest efficiency is 0.8, with 0.38 V and 3 μW cold-start gates. Ready/off voltages are 2.7/2.3 V; maximum capacitor voltage is 3.3 V, rating 5.5 V and ESR 0.3 ohm. The managed output is 1.8 V with 0.3 V dropout. Off/sleep power is 1/5 μW; boot and storage-side TX costs are 100 and 130 μJ. Transmission uses the existing ideal atomic wire/CRC and receiver abstraction, not a qualified RF link.

The O addition declares 0.5 μW powered standby, 1 μJ per wake, 20 μJ per successful RMS sample and 5 μJ per committed write. The 20 μJ value borrows the numerical A/R temperature-sample cost; it does not qualify a vibration sensor. A write is charged when a record is committed, not at every poll. NVM preserves committed state and the pending queue but provides no sensing energy. Powered samples use a declared zero-order hold; failed samples and brownouts reset continuous-confirmation timers. Energy integration is piecewise and observations are compressed into sampling runs. These are explicit digital assumptions with physical qualification UNKNOWN [4].

2.3 Candidate, controls and public observations

All three episodes share the same source, initial state, components, link, service predicate and hardware. The legal candidate contains only typed capacitance, sample period, thresholds, cache, retry/backoff and program choices. The fixed periodic control takes one RMS snapshot every 60 s and uploads it without an event confirmation state machine. The rich-state control uses a one-second sampled start/stop state machine, 120 s active keepalive, NVM queue, one-second retry and zero TX backoff. Each fixed control is its episode's sole legal design; the candidate may check at most two distinct designs under the same registered total allowance.

Every initial O0 episode has a ceiling of 24 provider requests, 40 tool calls, 1200 s and two distinct design exposures. Participants must actually inspect, check, judge and submit. Source records carry hashes and

unit maps. The common public-response projection replaces every array longer than twelve entries with its full count, canonical full-value hash, first six and last six entries. The complete raw digital response remains host-sealed, and public feedback contains no reference truth or host service label. Thus an evaluator's use of the full response is separate from what the participant could inspect [5]. All three episode receipts report the `deepseek-flash` provider alias; the protocol leaves pinned weight identity UNKNOWN.

3 Frozen procedures, selection and final information

H0 is the unchanged base design procedure: read the fixed task and available sources, choose a legal D, actually check it, then submit a source-bound design and judgment while retaining failures and UNKNOWN. G0 is the fixed parent improver. A single actual proposal episode forms G1 from the frozen parent material and retained public history. These are changes to instructional procedures; no model weights, scientific semantics or device parameters are trained by the G-formation step.

G1's recorded proposal calls for source-parameterized H text, capability and budget checks before spending an attempt, separation of achieved delivery from post-terminal decay, and retrospective rehearsal of quantitative predictions on their own source episode. It distinguishes arithmetic/transcription errors from substantive design failures and retains no-update as an option. This describes the proposed procedure; it does not establish that every consumer follows it or that any rule is a novel or necessary mechanism [6].

Three lineages each generate two H proposals under G0 and two under the same shared G1, for twelve actual H proposals. Each admitted descendant is consumed on the fixed pulse-fast and volatile-reset selection tasks. H0 supplies the corresponding control in each lineage. Thus selection contains 24 descendant cells and six H0 cells. The registered ranking prefers more complete-service passes, then fewer nominal judgment disagreements, then fewer charged requests, then the lower descendant index. Proposal costs and both selection attempts enter the request criterion; unselected sibling costs remain in the pipeline. All 30 effective selection cells pass with no nominal judgment disagreement. All six choices are frozen before final exposure [7].

Each selected descendant and H0 is then consumed by fresh design episodes on three frozen final inputs: a stretched-port-window transient task, a half-power C-lite query derivative, and a new O waveform. The C-lite branch is the existing single-query digital abstraction, not a new RF or physical environment. The final designer can inspect the complete current task and acquire its sources, including the new O waveform. Formation and selection do not see these final inputs, and final feedback cannot trigger reselection. O0 is excluded from G/H formation and selection. Array projection is the same in all arms.

Later D episodes have at most 12 provider requests, 40 tool calls, 800 s and two distinct designs. Each H proposal has an eight-request/400 s ceiling. The one G1 formation has an eight-request/400 s ceiling; its actual three requests are charged once and included in G1 pipeline cost. The nine final cells per arm represent three fixed tasks consumed in three descendant-lineage contexts, not nine independent task families [5, 10].

4 Observed O0 behavior

4.1 Event success is shared with the rich-state control

Table 1: Three actual O0 development episodes. Matched counts concern eight transitions on one task. Candidate and rich-state share one digital response.

Arm	Poll (s)	Matched	Host O	Self-judgment	Requests
Candidate	1	8/8	PASS	UNKNOWN	8
Fixed periodic	60	0/8	FAIL	FAIL	5
Rich-state	1	8/8	PASS	PASS	5

Candidate and rich-state submit identical designs: 15 mF, one-second sampling, thresholds 0.05/0.01 g, NVM, one-second retry, zero backoff and the confirmed-event program. They share the same execution key and response hash. The external event evaluator passes all eight transitions, energy and receiver checks, with no

missing keepalive or idle false event. The shared response contains 18 accepted packets. Two fresh model submissions of one shared digital response do not supply two independent realizations [3].

The periodic control's energy and receiver checks also pass, with 348 accepted packets. Its messages are periodic snapshots, which provide neither the required start/stop records nor active-state keepalive coverage. It matches zero of the eight transitions and fails event service. The contrast shows that energy feasibility and accepted packet count do not by themselves satisfy this event contract. Because the fixed rich-state control already passes, O0 establishes no additional event-service benefit for the candidate interface. The candidate also uses eight provider requests, versus five for each control; these single-episode counts supply no efficiency advantage or statistical comparison [1].

4.2 Retained checked alternative and timing sensitivity

The actual candidate checks one-second sampling first, then 1.5-second sampling, and finally submits the one-second design. Both checked responses are retained in its exposure and cost history. In the 1.5-second check, the first start record is observed at 4204.070423 s and stamped 4204.139063 s, against the external confirmation time 4203 s. Its timestamp error is 1.139063 s, exceeding the frozen 1 s tolerance even though energy and delivery observations alone can appear adequate. This observed timing violation explains retention of the one-second design; it is a checked alternative within the same episode, not another model repetition or a later repair to the accepted design [3].

4.3 Participant UNKNOWN and the host event PASS remain distinct

The candidate's submitted service judgment is UNKNOWN. Its original explanation identifies unobserved dark intervals after brownout, the absence of an O gap message to bracket those intervals, inherited A/R first-receive and dark-report metadata, and unavailable source/hardware calibration. Specifically, the inherited metadata request reception by 1800 s and dark reporting every 600 s; the first confirmed O event occurs much later. The participant therefore declines broader acceptance. Its complete original explanation is preserved verbatim in the manuscript binding as participant testimony, including its own numerical interpretations [2, 3].

The registered O evaluator instead asks whether the required start, active keepalive and stop history was delivered under the nominal energy/receiver model. It returns PASS, and the receipt retains a nominal predicate-disagreement flag. The participant's physical and broader-reporting UNKNOWN is not a physical self-judgment error. Both scopes are retained. Neither this disagreement nor the event PASS resolves the inherited broader reporting contract. Earlier A/R dark NO_GO and FIFO FAIL remain historical failures; O0 supplies no whole-product acceptance.

The one-second observations include dark-period brownouts near 8766.18 and 19587.64 s. The required active intervals do not overlap the subsequent unobserved gaps. Successful O0 event coverage therefore does not establish uninterrupted six-hour sensing or successful detection in every possible dark window. The next waveform must be evaluated as a separate frozen task rather than inferred from the O0 final state. Physical qualification remains UNKNOWN for all three arms.

5 Final results with every cell retained

5.1 Two digital tasks saturate; the new O task separates outcomes

Table 2: Final P/F/I counts: PASS, submitted FAIL and INCOMPLETE. The denominator is six digital cells plus three O cells per arm; all 27 remain included.

Arm	Two digital tasks	New O waveform	Total
H0	6/0/0	1/1/1	7/1/1
G0	6/0/0	0/1/2	6/1/2
G1	6/0/0	2/0/1	8/0/1

Both reserved digital inputs have three PASS per arm, for 18/18 final cells across those two inputs. All final submitted designs pass their energy gate. On the new O waveform, H0 has one PASS, one submitted FAIL and one INCOMPLETE; G0 has zero, one and two; G1 has two, zero and one. The complete 27-cell totals are H0 7/1/1, G0 6/1/2 and G1 8/0/1. Removing incomplete episodes would change the question and inflate the apparent completion rate [9, 10].

The final per-lineage pass vectors, in lineage order, are H0 $(2, 3, 2)$, G0 $(2, 2, 2)$ and G1 $(3, 2, 3)$. G1's 8/9 is a descriptive result from three descendant-lineage contexts conditional on one shared formation. There is no independent replication of G1 training, no large task sample, and no promotion of the default H0 or improver. Strong selection performance on two tasks also does not establish transfer to a different event-time alignment.

5.2 Encoded event time is stricter than event matching

Table 3: All nine new-O final cells. Every submitted row has energy PASS, eight matched events and 18 accepted packets. Events P/8 counts individually passing events, not matches. Dash denotes no submission.

Lineage-arm	Poll (s)	Retry (s)	Events P/8	Max error (s)	Service
L1-H0	1	1	6/8	1.103	FAIL
L1-G0	–	–	–	–	INCOMPLETE
L1-G1	0.75	0.25	8/8	0.728	PASS
L2-H0	0.75	5	8/8	0.728	PASS
L2-G0	–	–	–	–	INCOMPLETE
L2-G1	–	–	–	–	INCOMPLETE
L3-H0	–	–	–	–	INCOMPLETE
L3-G0	1	0.25	6/8	1.103	FAIL
L3-G1	0.75	1	8/8	0.728	PASS

All five O submissions match eight events, receive 18 accepted packets and pass energy checks. Only three pass complete event service. These three use 0.75 s sampling; the two one-second submissions each pass six individual event checks and fail two stops with `WRONG_EVENT_TIME`. The three successful designs include H0 as well as G1. Retry periods also differ, so this is an observed design association, not a controlled one-factor sampling ablation or an exclusive G1 mechanism [9].

Table 4: The two failed stops in each one-second submission. These same times occur in both failed designs, whose retry values and execution keys differ. The error is computed from encoded timestamp, not reception time.

Confirmation (s)	Reception (s)	Encoded time (s)	Error (s)
12077.625	12078.588	12078.629	1.004
17043.625	17044.588	17044.728	1.103

Both failed stop receptions occur less than one second after the external confirmation instant. Nevertheless, their encoded timestamps exceed the one-second error bound by 0.004 and 0.103 s. Sampling phase and the modeled clock's timestamp accumulation therefore have to be checked together. Correct matching, prompt reception, an empty pending queue and energy feasibility cannot substitute for the encoded-time predicate. The 0.75 s submissions have maximum encoded-time error 0.728 s on this waveform. Their observed success is specific to the frozen digital input and its timing, not a physical robustness guarantee.

The two submitted final O failures each self-report PASS; their nominal event-service disagreements are retained. These differ from the original O0 candidate's broader UNKNOWN: in the final failures the host identifies a specific encoded event-time violation within the submitted nominal contract. Missing submissions supply no service self-judgment at all.

5.3 Four token-censored episodes remain in the denominator

The incomplete final episodes are L1-G0, L2-G0, L2-G1 and L3-H0 on O. Each last SDK assistant response has stop reason `length`, with 32,768 output tokens and 32,768 reported reasoning tokens, and no final submission. This is a response-level metadata observation; it is not a six-hour device failure, an energy FAIL, or evidence that the whole episode used only that many tokens. The four episodes are not retried. Their requests and costs remain charged, and the outcome remains INCOMPLETE rather than a conditional success or an omitted row [9, 10].

6 Host cache incident and explicit protocol continuation

The later selection stage encounters a host representation error: historical `observations.json` is loaded without its separately sealed `trace.csv`, while assessment requires the trajectory. Three selection attempts raise `KeyError:'trace'`; a fourth attempt is truncated by the hold. These attempts are apparatus failures or censoring, not negative G0/G1 method outcomes. The frozen hold preserves all four records and their costs: 21 provider requests and USD 0.1318473 in settled upper-bound estimates [12].

The explicit amendment limits repair to reconstructing the already sealed historical trace and linking four named one-off recovery attempts. Qualification passes exact-input historical assessment, preservation of compressed raw and all five previously valid selection results, and rejection of missing trace, wrong trace hash, wrong input and wrong world. These checks use existing responses with zero new provider requests or simulations; they establish a cache decoding boundary, not method success [14].

Version 1.1 retains the original methods, tasks, scoring and request/time/dollar budgets, while preserving every original attempt. Formal selection accounting uses the original proposal and two effective selection attempts; incident overhead remains separately itemized and included in full arm/global pipeline cost. The protocol retains 73 logical slots with a ceiling of 77 actual episodes after four named recoveries. Request, summed episode-time and settled upper-bound cost ceilings remain 860, 54,400 s and USD 100. These are ceilings, not observed final totals. The amendment preserves the five valid selection responses and twelve H proposals rather than silently replacing them [13, 5]. The four named authorized recovery attempts all close with PASS. They resolve the four canonical slots without rebuilding G/H or erasing the incident history [15]. Effective selection consequently has 30 cells, while its full accounting contains 34 actual attempts.

7 Complete accounting and interpretation

Table 5: Full pipeline accounting from the authoritative analysis. Selection includes the four original host incidents and four recoveries. USD denotes fixed-rate provider upper-bound estimates; wall is summed episode time.

Stage	Attempts	Requests	USD	Wall (s)
O0 development	3	18	0.1579020	240.4
G1 formation	1	3	0.0269157	38.7
All H proposals	12	43	0.4491399	862.9
Selection, all attempts	34	173	1.2049959	2028.0
All final cells	27	130	1.3377693	2781.6
Total	77	367	3.1767228	5951.6

The full pipeline records 367 requests, all SETTLED, with no new UNKNOWN usage, USD 3.1767228 in fixed-rate settled upper-bound estimates and 5951.639484 s of summed episode wall time. This includes the one G1 formation, all twelve H proposals, eliminated siblings, selection, final episodes, four original host incidents, four recoveries and token-censored responses. The incident subset is 21 requests, USD 0.1318473 and 216.654733 s; it is already in the totals and must not be added again [10].

Full G0 pipeline cost is 135 requests and USD 1.2585825; full G1 cost is 138 requests and USD 1.2447618, including its formation. H0 selection plus final consumption costs 76 requests and USD 0.5154765, and O0

costs 18 requests and USD 0.1579020. These four buckets sum to the global total. G0 and G1 pipeline costs include both proposed descendants rather than only the selected child. Provider aliases and registered fixed rates bound an estimate, not a provider invoice or the full end-to-end research cost. Codex assistance and historical acquisition remain unpriced.

O0's scope is one development input and an explicitly assumed digital apparatus. The results do not qualify a photovoltaic device, PMIC transient model, vibration sensor, physical clock, NVM implementation or RF receiver. They also do not erase prior A/R reporting failures. A/R NO_GO, dark FAIL, strict FIFO FAIL and product NOT_ACCEPTED remain preserved. The independent O event gate does not cover every inherited A/R metadata, reporting or dark-observation requirement. Physical qualification remains UNKNOWN throughout.

The study establishes a finite separation: original O0 can be served by the same design as a strong fixed control; a different waveform exposes encoded-time violations despite successful energy and event matching; and procedural completion can fail through response truncation. It supplies neither evidence of an interface advantage beyond rich-state nor a stable recursive-improvement mechanism. The unchanged H0's own successful 0.75 s submission and the single shared G1 formation constrain that interpretation.

8 Reproducible manuscript and evidence limits

`render_phase4.py` verifies the authoritative closed handoff and its finite source list, cross-checks actual and final rows, frozen choices, accounting and per-event timestamp arithmetic, then generates this fragment. It also preserves the original O0 participant explanation verbatim in `phase4-binding.json`. It runs no model, scientific simulator or service evaluator. The source-preservation receipt records all six choices frozen before final exposure, unchanged historical prefixes and preserved old UNKNOWN billing records [11].

The paper-final receipts bind both TeX sources, renderer, shared layout/logo, compiled PDF, fonts, extracted text and actual page-by-page visual review. Build and rendering validate presentation and source binding; they do not replicate the scientific batch, establish hardware acceptance or replace the coordinator's registered repository verification. No scientific code, task, result, old manuscript, shared style or logo is changed for this paper.

References

[1] S0. Three registered actual O0 transfer episodes. `O0-transfer-results.json`, 20 October 2026.
[2] S0. First O0 result and retained claim boundary. `first-O0-result.json`.
[3] S0/Pi. Original O0 candidate, fixed-periodic and rich-state `episode-result.json` receipts, and candidate gateway check records. Exact paths and SHA-256 bindings appear in `phase4-binding.json`.
[4] S0/B. Original O0 case, declared digital components and versioned event evaluator. `o0-task.v1.json` and `phase4_interfaces.py`; the actual execution versions are source-bound.
[5] S0. Finite Phase 4 protocol, version 1.1. `protocol-v1.1.json`; original O0 results are preserved.
[6] S0. Actual G/H parent and procedure freezes. `parent-freeze.json` and `G1-freeze.json`.
[7] S0. Thirty effective selection results and all six frozen choices. `selection-complete.json`, `selected-methods.json` and `final-selection-freeze.json`.
[8] S0. Complete 77-attempt and 73-cell record. `actual-results.json`.
[9] S0. All 27 final cells, encoded event-time details and censored response metadata. `final-details.json`.
[10] S0. Descriptive task/arm counts and complete pipeline costs. `analysis-summary.json`.
[11] S0. Final frozen-source, accounting and history readback. `preservation-final.json`; manuscript transfer in `paper-handoff.json`.
[12] S0. Historical cache exception and stopped-attempt record. `host-error-hold.json`.
[13] S0/review. Authorized historical-trace reconstruction and four explicit recovery links. `amendment-registration.json`.
[14] S0. Qualification of cache reconstruction and negative provenance controls. `cache-repair-qualification.json`.
[15] S0. Four approved recovery attempts linked to retained host incidents. `recovery-results.json`.